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- 1) b) Data and a pointer to the next node.
- 2) c) Null
- 3) b) self-referential structure.
- 4) b) head == Null
- 5) b) newnode $\rightarrow$ next = head; head = newnode
- 6) b) temp = head; head = head $\rightarrow$ next;
free(temp);

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7) Traversing from tail to head.

- 8) b) n
- 9) c) copying the data of $p \rightarrow$ next into p,
then deleting $p \rightarrow$ next.

10) b) while ($p \rightarrow$ next $\neq$ null)

11) b) $p \rightarrow$ next == Null

12)

13) c) Tree

14) b) Exactly one node.

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15) b) $p = (\text{struct Node}^*) \text{malloc}(\text{sizeof}(\text{struct Node}))$

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- 16) b) Avoid special case handling for insertion and deletion at the head.
- 17) c) The first node.
- 18) b) It does not support random (indexed) access.
- 19) b) Loss of access to all nodes, causing a memory leak.
- 20) b) A slow pointer and a fast pointer.
- 21) c) ^{The} last node.

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- 22) b) Nodes may be scattered anywhere in memory.
- 23) #include <stdio.h>
#include <stdlib.h>
struct Node {
int data;
struct Node* next;
};

24) 10 → 20 → 30 → 40 → Null

Output:- 30 → 50 → 70

15)

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```
25. void insertEnd ( struct node * head, int x ) {  
    struct node = t = (struct node) malloc  
        (size of (struct node));
```

```
    t -> data = x;
```

```
    t -> next = null;
```

```
    if (*head == null) {  
        *head = t;  
        return;
```

```
    }  
    struct node * p = *head;
```

```
    while (p -> next != Null)
```

```
        p = p -> next;
```

```
    p -> next = t;
```

```
}
```

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```
6. struct node * reverse
```

```
    (struct node * head) {
```

```
    struct node * prev = Null;
```

```
    struct node * curr = head;
```

```
    struct node * next = null;
```

```
    while (curr != null) {
```

```
        next = curr -> next;
```

```
        curr -> next = prev;
```

```
        prev = curr;
```

```
        curr = next;
```

```
}
```

```
    return prev;
```

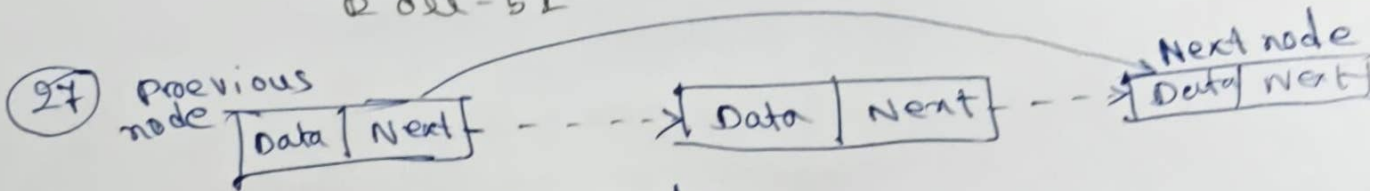
```
}
```

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pointer adjustment
 $\text{prev} \rightarrow \text{next} = \text{del} \rightarrow \text{next};$
 $\text{free}(\text{del})$

27) $5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{Null}$
 $7 \rightarrow 5 \rightarrow 8 \rightarrow 12 \rightarrow 3 \rightarrow \text{Null}$
 $7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{Null}$
Final $\rightarrow 7 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 9 \rightarrow \text{Null}$

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